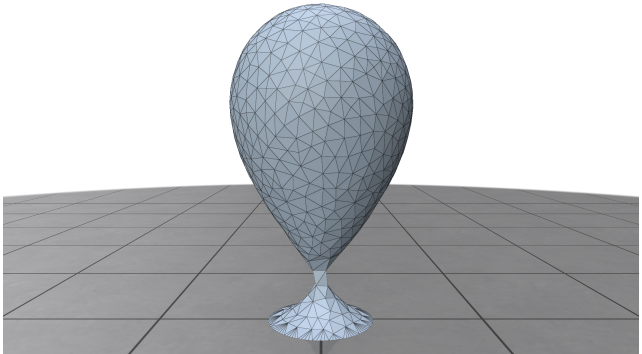
Universität Bremen

EFDC2
2ND EUROPEAN FLUID DYNAMICS CONFERENCE
26TH - 29TH AUGUST 2025
DUBLIN, IRELAND

Bubble detachment from heterogeneous surfaces

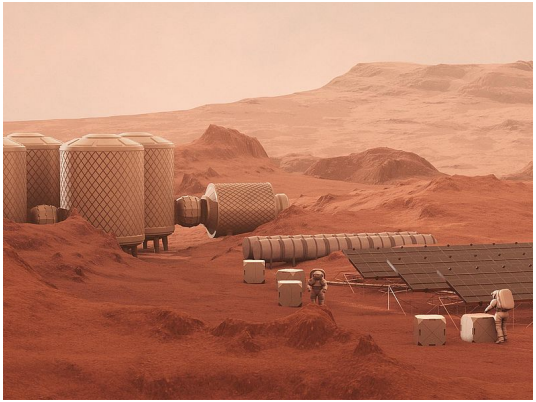
27 Aug 2025, EFDC2, Dublin, Ireland.

Ianto Cannon, Stefan Endres, Lutz Mädler, and Marc Avila



The need for hydrogen and oxygen

- Hydrogen and oxygen are needed as fuel for interplanetary travel.
- Less energy is needed to escape Earth's gravity if the rockets are refuelled on a lunar base
- Humans on the Moon and Mars will need supply of oxygen
- Ice is found at the poles and in craters on the Moon and Mars
- Solar photovoltaics can be used to generate electricity



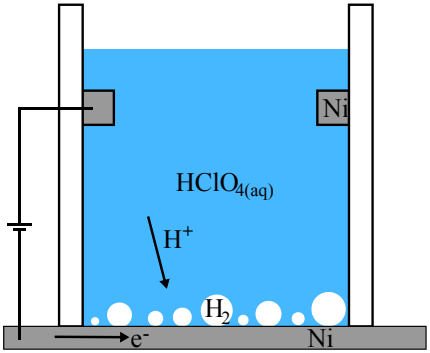
Joris Wegner, University of Bremen

Electrolysis of acidic water

- Hydrogen bubbles form at the anode

$$2H^+_{(aq)} + 2e^- \rightarrow H_{2(g)} \qquad (1)$$

- Attached bubbles obstruct the motion of H^+ ions, reducing efficiency



Demirkr, a., Wood, J. A., Lohse, D., & Krug, D. (2024). *Langmuir*, (39)

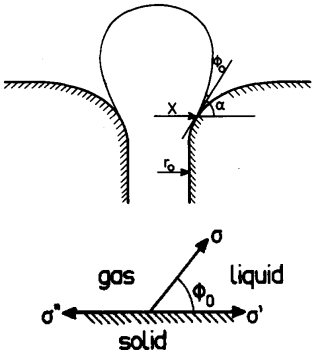
Aims

Question

What size are the bubbles when they detach from the electrode?

Bubble formation

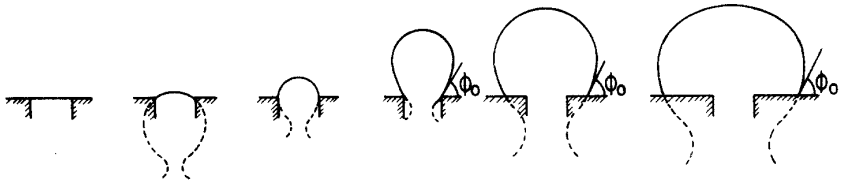
- Bubbles nucleate inside cavities on the electrode surface.
- The contact angle ϕ_0 between the three phases depends on their relative affinities.



Chesters, A. K. (1978). *International Journal of Multiphase Flow*, (3)

Pinned and spreading bubbles

- At first, the contact line lies on the curved edge of the cavity as the bubble grows.
- The contact line remains on the cavity as long as the angle of the bubble foot with the horizontal is more than ϕ_0
- This is known as the **pinned** state.
- If the angle of the interface decreases to ϕ_0, the bubble spread across the electrode.
- This is known as the **spreading** state.



Chesters, A. K. (1978). *International Journal of Multiphase Flow*, (3)

Bubble shape

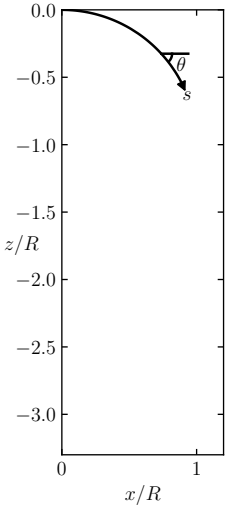
- The curvature of the interface is given by the Young-Laplace equation,

$$\sigma\left(\frac{1}{R_1}+\frac{1}{R_2}\right)=p_{out}-p_{in} \tag{2}$$

- Assume axial symmetry,

$$\sigma\left(\frac{d\theta}{ds}+\frac{\sin\theta}{x}\right)=(\rho_{in}-\rho_{out})zg-\frac{2\sigma}{R} \tag{3}$$

Bashforth, F., & Adams, J. C. (1883). Cambridge University Press

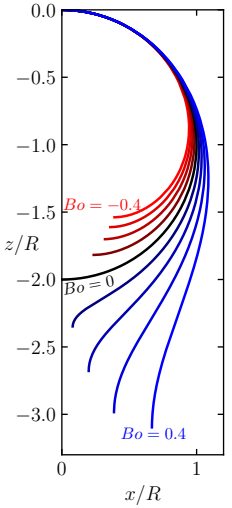


Bubble shape

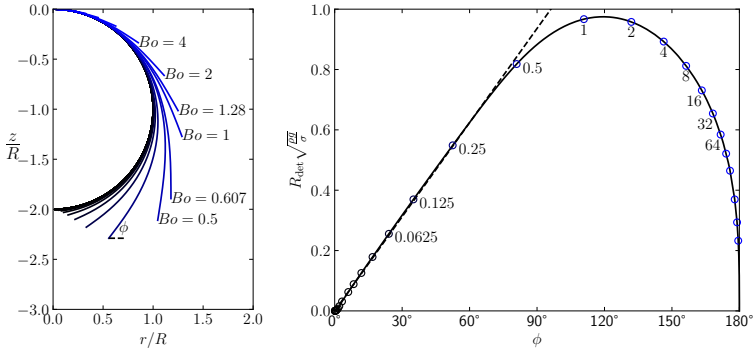
- Group dimensional variables in the Bond number $\mathrm{Bo} = \frac{(\rho_{in}-\rho_{out})gR^2}{\sigma}$;

$$\frac{d\theta}{d(s/R)} + \frac{R}{x}\sin\theta = \mathrm{Bo}\frac{z}{R} - 2. \tag{4}$$

Bashforth, F., & Adams, J. C. (1883). Cambridge University Press



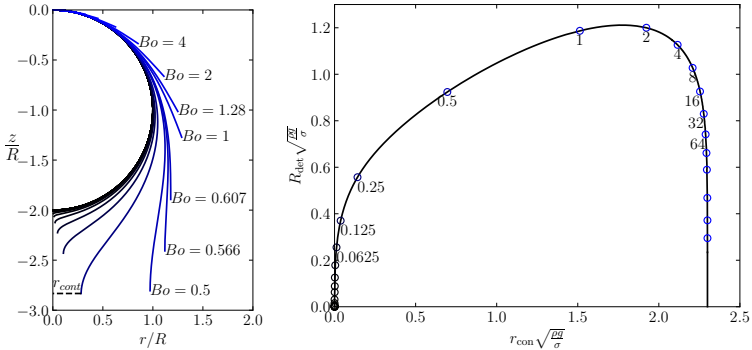
Detachment size of spreading bubbles



- For each Bond number Bo, we find the largest bubble with a given contact angle ϕ.
- The largest spreading bubble has a contact angle of 121° and radius of 0.97 capillary lengths.
- The dashed line $R_{\mathrm{det}}\sqrt{\frac{\rho g}{\sigma}} = 0.104\phi$, is the result of Fritz, 1935.

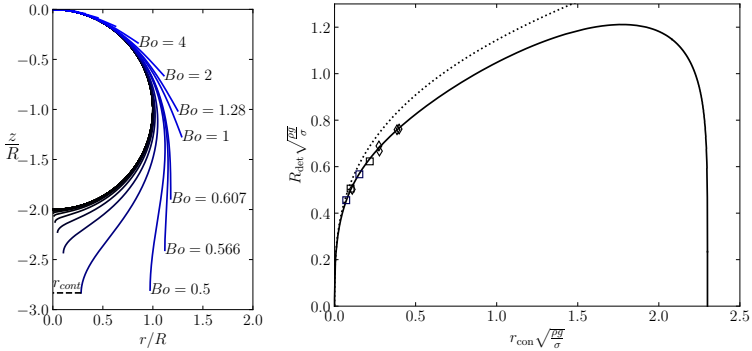
Universität Bremen

Detachment size of pinned bubbles



- For each Bond number Bo, we find the largest bubble with a given cavity radius r.
- The largest possible bubble has $R_{\mathsf{det}}\sqrt{\frac{\rho g}{\sigma}} \approx 1.2$.

Detachment size of pinned bubbles

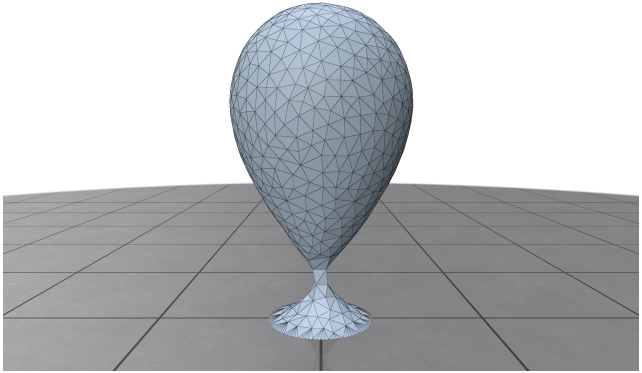


- Our results agree with the carbonated water experiments of Mori and Baines, 2001 and the boiling experiments of Lesage and Marois, 2013.
- We measure a maximum possible cavity radius of 2.40, agreeing with the stability analysis of Pitts, 1974.

Aims

Question

Can we predict the shape of generic 3D bubbles?

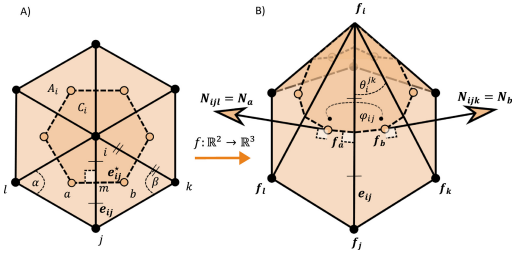


Surface tension force

- Triangulate the interface so that vertices lie on the surface.
- Compute the total curvature at each vertex using discrete differential geometry.

$$\widehat{H\mathbf{N}A_i} = \frac{1}{2} \sum_{j \in st(\mathbf{f_i})} \left(\cot \alpha_{ij} + \cot \beta_{ij}\right) \left(\boldsymbol{f}_i - \boldsymbol{f}_j\right). \tag{5}$$

- Apply a surface tension force $\boldsymbol{S_i} = \sigma \widehat{H\mathbf{N}A_i}$.
- Endres, S. C., Avila, M., & Lutz, M. (2022). *Chemical Engineering Science*



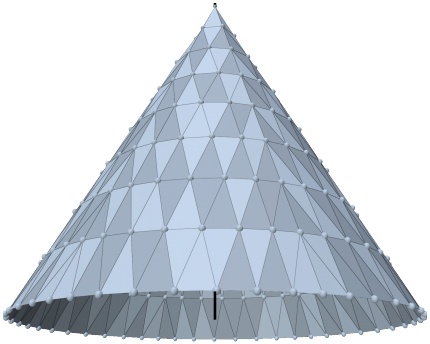
Pressure forces

- Initialise the bubble as a cone with volume V_0 and base radius R_{foot}
- Outer pressure assumed to be hydrostatic.

$$p_{\text{out}} = \Delta\rho g z \tag{6}$$

- Inner pressure enforces volume conservation.

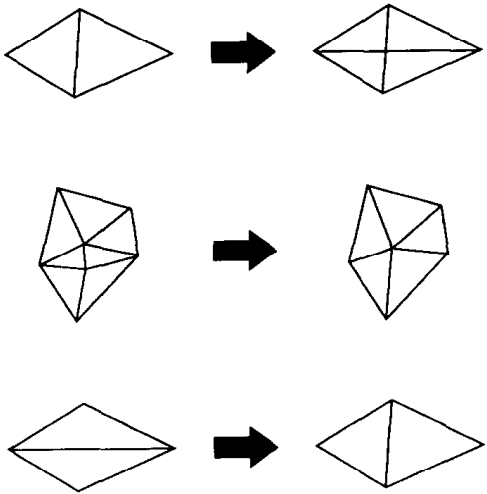
$$p_{\text{in}} = \frac{2\sigma}{R}\left(\frac{V_0}{V}\right)^5 \tag{7}$$



Remeshing

Use the method described in Unverdi, S. O., & Tryggvason, G. (1992). *Journal of Computational Physics*, (1);

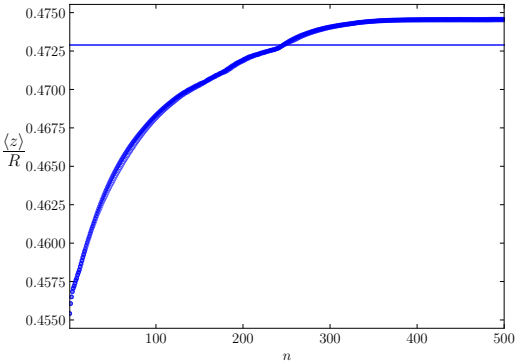
- Split edges longer than $0.2R$
- Merge edges shorter than $0.1R$
- Reconnect diagonals if it shortens them by more than a factor of 1.5.



Minimising free energy

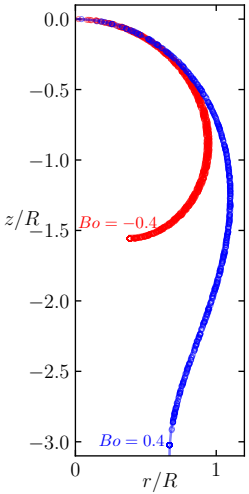
Use the Adams Bashforth method to find the locations $\mathbf{x_i}$ where resultant forces $\mathbf{F_i}$ on the vertices are zero;

$$\mathbf{x_i^{n+1}} = \mathbf{x_i^n} + \alpha\frac{3}{2}\mathbf{F_i^n} - \alpha\frac{1}{2}\mathbf{F_i^{n-1}} \qquad (8)$$

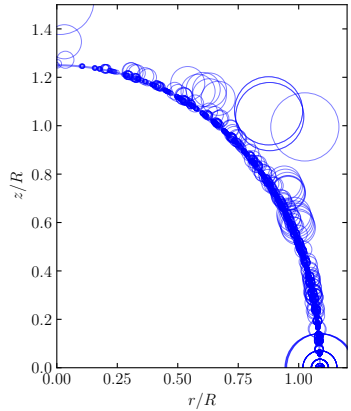


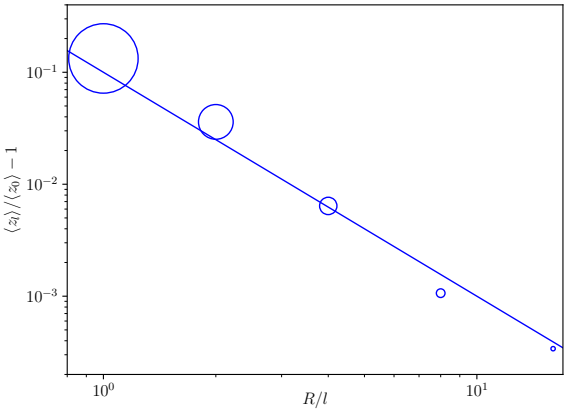
Validation: bubble shape

For a fixed contact line, we can replicate the bubble shape for a range of Bond numbers, $\mathrm{Bo} = \dfrac{\Delta\rho g R^2}{\sigma}$.



Validation: grid dependence





The error in the height of the centre of mass has a quadratic dependence on the grid size,

$$\frac{\langle z_l \rangle}{\langle z_0 \rangle} - 1 = 0.1 \left(\frac{R}{l} \right)^{-2}. \tag{9}$$

Conclusions

- The largest detaching bubble has a radius of 1.2 capillary lengths.
- The largest cavity that can support a bubble has a radius of 2.4 capillary lengths.
- We developed a discrete differential geometric formulation with second-order convergence.

Thank you!

https://ianto-cannon.github.io/

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